# WRITTEN EXAM OF TMM

**Legend of Symbols[8]:**
TM = tempered martensite, B = bainit
BL = lower bainite, P = pearlite
RA = residual austenite, F = ferrite,
A = austenite, BU= upper bainite;
M = martensite
DBTT= ductile-brittle transition temperature
÷ division operator
/ separator
$T_A$ = austenitizing temperature

**1. It is FALSE that high energy in GBs promotes:**

| | | | |
|---|---|---|---|
| (a) dislocation blocking | (b) corrosion | (c) nucleation of new phases | (d) easy crossing of dislocations |

**2. Slip systems in metallic crystals can be accounted by:**

| | | | |
|---|---|---|---|
| (a)disloc. density | (b) vacancy concentration | (c) Schmid's factor | (d) Hall-Petch law |

**3. Toughness of metals is NOT markedly affected by:**

| | | | |
|---|---|---|---|
| (a) high temperature | (b) solphur segregations | (c) inclusion content | (d) grain size |

**4. Extensive solid solution in alloys is attained when**

| | | | |
|---|---|---|---|
| (a) diffusion activation energy is high | (b) dislocation density is large | (c) applied stress is large | (d) electronegativity difference is low |

**5. In the Fe–$Fe_3C$ phase diagram the eutectic temperature is:**

| | | | |
|---|---|---|---|
| (a) 1127°C | (b) 1147 °C | (c) 727 °C | (d) 1494 °C |

**6. Increase in A1 temperature in steels depends on**

| | | | |
|---|---|---|---|
| (a) carburizing elements | (b) grain size of ferrite | (c) amount of pearlite | (d) heat treatment |

**7. The eutectic in theFe-$Fe_3C$ phase diagram is a mixture of**

| | | | |
|---|---|---|---|
| (a) cementite + liquid | (b) α + cementite | (c) α + γ | (d) γ + cementite |

**8 imate the proportion of pearlite for the $C_o$-alloy at To by nating the %C as suggested in the figure**

| | | | |
|---|---|---|---|
| 0.8 | (b) ~0.50 | (c) ~0.4/7 | (d) ~0.8/7 % |


**9. The meaning of C-temperature and R domain (C/R) in figure is**

| | | | |
|---|---|---|---|
| (a) $A_{cm}$ cementite | (b) $A_3$/ α+$Fe_3C$ | (c) $A_1$ /α+$Fe_3C$ | (d) Acm /α+ $Fe_3C$ |

**10. The microstructure of tempered martensite is made of:**

| | | | |
|---|---|---|---|
| (a) $Fe_3C$+RA+α | (b) BL+RA | (c) $Fe_3C$+BL | (d) $Fe_3C$+α |

**11. How does alloying of Ni affect the TTT curves (figure)**

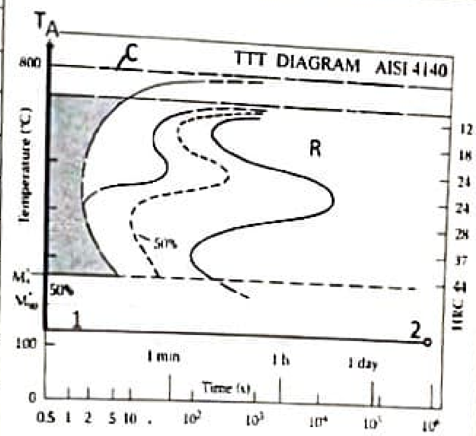
| | | | |
|---|---|---|---|
| (a) decrease $M_{90}$ | (b) Increase $M_{90}$ | (c) Shift left down | (d) increases $T_A$ |

**12. It is FALSE that at low temperatures of the steel TTT diagram:**

| | | | |
|---|---|---|---|
| (a) grain size decreases | (b) hardness increases | (c) nucleation rate increases | (d) diffusion decreases |

**13. Estimate the % of cooling products in 1-2 transformation in figure**

| | | | |
|---|---|---|---|
| a) 90%M + 10%AR | b) 90%M + 10% TM+BL | c) 100% M | d) 100% TM+BL |




**14. The exponent of Ludwik-Hollomon's power law is NOT related to:**

| | | | |
|---|---|---|---|
| a) strength | b) crystal structure | c) plastic strain | d) yield stress |

**15. Nitriding of steels DOES NOT involve:**

**(a)** surface diffusion **(b)** post-heat treatment **(c)** compressive residual stresses **(d)** γ' phase formation

**16. Which steel, among AISI 1040, 4140 and 4340, has largest hardenability after Jominy test?**

**(a)** AISI 1040 **(b)** AISI 4140 **(c)** AISI4340 **(d)** all same

**17. The purification reaction during production of pig iron in the blast furnace is:**

**(a)** $CaCO_3 \rightarrow CaO + CO_2$ **(b)** $CO_2 + C \rightarrow 2\,CO$ **(c)** $C + O_2 \rightarrow CO_2$ **(d)** $3\,CO + Fe_2O_3 \rightarrow 2\,Fe + 3\,CO_2$

**18. Which is the most effective degasing method to eliminate H gas in liquid steel**

**(a)** AOD **(b)** VOD **(c)** BOF **(d)** Electric Arc furnace